

Examiner reports

Q1.

In part (a), the basic principles were well recalled, and the majority correctly handled the minus sign. Extra terms were not penalised.

In part (b), a proportion ignored the instruction “Using your expansion”, and so scored no marks. A common error was to use 0.994, or 0.006 for x . As in part (a), those who had used more terms to obtain 0.976215 were not penalised on this occasion.

Q2.

This question proved challenging for many students. Many were uncertain how to proceed and seemed unsure about the term ‘coefficient’ and were unable to work with the required „C, terms. A significant number of students attached the 1.5 to the wrong side of the equation. It was pleasing to see many students who were unable to complete an algebraic solution successfully resort to using a numerical approach which often generated the required $n = 11$. It was surprising that more students did not resort to using the equation solver on their calculator to solve their algebraic equation.

Q3.

This question discriminated well between lower and higher attaining students. The start of the question was accessible to most students. The later parts were more demanding and only the higher attaining students were successful.

Part (a) was completed successfully by a high proportion of students. Many selected a correct approach and manipulated the expression so that the binomial expansion could be applied. The most common error was to extract a factor of 2 instead of $\frac{1}{2}$ from the square root.

Those students who spotted the connection to the first part often did well on part (b). Full marks were available even if the first part of the question had gone wrong, provided $-x^3$ was correctly substituted into their three-term expansion. Sign errors were the most frequently seen mistakes.

In part (c) students could still achieve credit even if their previous answers had gone wrong. Simply substituting their three-term answer to part (b) to form the integral would achieve the first mark.

Very few students demonstrated the insight required for (d)(i) and this question was intended to distinguish the most able students. Those who attempted an answer often talked about the shape of the curve or the use of trapezia. Those who realised that the key was that all the terms were positive often score full marks.

The idea being tested in (d)(ii) was standard, but it was made more difficult by the context. Students were not directed to find the range of validity for their expansion, but had to make the connection for themselves. The first mark could be achieved provided they used their expansion from part (b).

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Both parts of question (d) required explanation. Generally, this was not done well. Students needed to state a key fact which must be true/false and then explain the implications of this.

Q4.

This was a routine question, which was answered correctly by a high proportion of students. The most frequent incorrect selection was evenly split between 42 and 21, but 4 was not far behind. This may suggest that students who did not know the answer guessed rather than being an indication of a particular misconception.

Q8.

Part (a) was generally answered correctly although all the usual errors were seen in this simple type of expansion. A few students did not simplify the terms in their correct expansion in part (a) but credit was awarded retrospectively if the correct simplified expansion was seen in the later part (c). Many students understood what was required in part (b) but a very common wrong answer, $1 + 2x + 7x^2 + 14x^3$, was seen due to the errors $\left(\frac{x}{4}\right)^2 = \frac{x^2}{4}$ and $\left(\frac{x}{4}\right)^3 = \frac{x^3}{4}$. In general, only the better students were able to find the three relevant products which led to the coefficient of x in the expansion for part (c) being 30.

Q9.

Overall, this question was answered well and was a good source of marks for most students. Over half the entry scored 10 or more of the 11 marks available.

Part (a). The vast majority of students answered correctly, showing confidence in finding partial fractions. The most common method was to substitute $x = 3$ and $x = -\frac{1}{3}$, although some students used simultaneous equations and others hybrid methods. Few mistakes were seen, although some students did manage to swap over the given coefficients A and B .

Part (b)(i). Many students approached this by writing out the full expression using their values of A and B and using the binomial expansions of both terms clearly and correctly to arrive at the correct answer. The common errors were not to take the 3 out of $(3 - x)$ correctly, if at all, and / or to get confused with the index -1, this sometimes being combined with the 2 (value of A) to become $\frac{3}{2}$, 6 or $\frac{1}{6}$. The other common error was to drop the minus sign in the expansion of $\left(1 - \frac{x}{3}\right)^{-1}$. Although most students made a clear attempt to combine the two expansions, some made sign errors. Very few students attempted to use the binomial theorem result from the formula book, or did not use the partial fractions and multiplied their expansions together with $(7x - 1)$, although a few correct solutions were seen using this method.

Part (b)(ii). Most students were aware that this question involved the range of validity of

the expansions, with many finding the range correctly but not relating this to the given value of 0.4; this didn't answer the question. Other students seemed to just make a guess, such as '*not enough terms have been considered*' or even '*because 0.4 is a decimal number*'.

Q10.

This question tested expansion of brackets and integration of negative exponents.

The majority of students used the binomial expansion or expanding brackets successfully in part (a). Typically in a(ii) they repeated their working from part (a)(i), replacing y with x^{-2} , rather than replacing it in their simplified expansion. Most students scored at least M1 for this part, however a significant number of students were unsuccessful in finding a correct expression for $(2 - x^{-2})^3$ and hence did not find the correct values of p and q .

Those students who did not use "hence" but used a correct different method to the one they used in a(i), could gain a maximum of 2 out of the 3 marks. Part (b) made use of the answer to a(ii) and those that had found values for p and q were generally able to correctly integrate their expression, $p + qx^{-4}$, and evaluate it using the given limits in a correct manner.

Q11.

The majority of the students showed competence with binomial expansions, with about two thirds of the entry scoring 5 or more of the 7 marks available.

- (a) Very few students failed to score at least the method mark here.

One relatively rare error was to forget to square the 6 in the coefficient of the x^2 term.

- (b) (i) Most students correctly left $\frac{6}{27}x$ in the brackets and took out the 27 to the correct power, even if later they wrote this as 3 rather than $\frac{1}{3}$. Most students chose to expand $(1 + \frac{6}{27}x)^{\frac{1}{3}}$ and were usually successful; those who chose to use the result from part (a) mostly didn't realise they had to use $\frac{1}{27}x$ and not $\frac{6}{27}x$ with the part (a) coefficients. Few students attempted to use the result from the formula book, and of those who did, many made an error in evaluating the coefficients. A few mistakenly tried to use ${}_nC_r$ with $n = -\frac{1}{3}$.
- (ii) This was not answered well and was often not attempted. Those who did find $x = \frac{1}{6}$ as the required value of x made an error in their arithmetic after substituting it, or failed to double the result to get the correct answer. Other students found a whole miscellany of values of x , which led to answers which

could not possibly be correct, as a quick check on the calculator would have shown. Some students just gave the result from their calculator.

Q12.

In part (a)(i), most candidates recognised that the transformation was a stretch, but the details of the stretch were not as well known. A common error was to state the scale

factor as $\frac{1}{6}$ rather than the correct scale factor. Part (a)(ii), finding a suitably simplified expression for $g(x)$, proved to be one of the least well answered parts on the paper. The most common wrong approach was to just subtract 3 from the expression inside the

bracket and simplify to get $\left(-2 + \frac{x}{3}\right)^6$. The expected error, replacing x by $x + 3$, was also seen quite regularly. More able candidates, who correctly replaced x by $x - 3$, did not always simplify the resulting expression correctly to an acceptable form. In part (b), most candidates successfully applied the binomial theorem as far as finding the binomial coefficients 15 and 20. A significant proportion of these candidates went on to reach the correct answers, though a number of others failed to identify the required coefficients

completely or to give $\frac{15}{9}$ in its simplest form. A significant minority could not cope with the additional powers of $\frac{1}{3}$.

Q13.

Most candidates had part (a) correct, although some made an arithmetic or sign error, most commonly in the x^2 term.

In part (b), most candidates showed they knew the conventional starting point of factoring out the 8, and did this correctly, with relatively few losing the $2/3$ power on the 8 or not

having $\frac{6}{8}$ or $\frac{3}{4}$ as the coefficient of x . Some made arithmetic slips, particularly in the x^2 term. Others got to a correct binomial expansion in spite of omitting some brackets from their working; this was condoned on this occasion.

In part (c), the common mistake was for candidates to equate $8 + 6x$ to 100, rather than

to 10, and to work with $x = \frac{46}{3}$, apparently not noticing that this gives a negative number as a result. However, a significant minority of candidates were able to see, or work out,

that the value of x they needed was $\frac{1}{3}$ and completed the question successfully.

Q14.

This question, which tested positive rational exponents and integration, was another good source of marks for many students, although for a large minority finding the square of $x^{\frac{3}{2}}$

proved to be more of a challenge than was hoped, with $x^{\frac{9}{4}}$ appearing far too often from otherwise promising students. The indefinite integration and evaluation of a corresponding definite integral were tackled with much more success, with many correct solutions seen.

Q15.

In part (a), most students got the first two terms of the expansion correct, but often there was a mistake in the x^2 term, with $4x$ becoming just x being the common error. Some students made arithmetic errors with 4^2 , by failing to actually square the 4, and others failed to simplify the binomial coefficient correctly. For part (b)(i), most students took out 4 as a factor correctly, although some followed this with

an error in interpreting $4^{-\frac{1}{2}}$. Others left x , rather than $\frac{x}{4}$, as the term in the bracket. Those who did have the correct bracket at this stage usually expanded correctly, although

some made sign errors, particularly $\frac{x}{4}$ instead of $\left(-\frac{x}{4}\right)$, whilst again some could not simplify the binomial coefficients correctly. A common error was with the sign on the x -term, which lost the final mark. Some lost the final mark for not multiplying their

expansion by $\frac{1}{2}$.

Expansions using the result from the formula book were rare, but usually were done correctly when attempted.

Relatively few students wrote down a correct inequality in part (b)(ii). Common errors were $x < 4$, $x > 4$, $\frac{x}{4} < 1$ or $x \neq 4$. Some students used the modulus symbol correctly, whereas others showed no understanding, writing expressions such as $|x| < -4$.

Many students understood what was required in part (c) and attempted to multiply their two expansions, some doing this efficiently and correctly, but others doing more work than necessary, as only terms up to x^2 were required. Many algebraic errors were seen. Some students just multiplied terms of the same power. Some wrote down one expansion divided by another and divided each term in the numerator by the corresponding term in the denominator, but often a quadratic expression just appeared with little or no working seen. Some students added or subtracted their expressions with no apparent attempt to multiply them. Some put a square root sign around their final answer, showing little understanding of what the question was asking.

Q16.

A majority of candidates gave the correct binomial expansion in part (a), although a common error amongst the less successful candidates was to write the third term as $-3x^2$ instead of $+3x^2$. In part (b), most candidates picked up the method mark for their expansion of $(1+y)^4$. However, it was not uncommon to see errors in collecting like terms, the most common of which were ' $6y^2 - 3y^2 = -3y^2$ ' or ' $6y^2 - 3y^2 = 9y^2$ '. It was disappointing to see a small minority of candidates either having expansions entirely in terms of x or having a mixture of terms involving x and y with no indication of any corrections despite the printed answer.

Part (c) was generally only answered correctly by the most able candidates who replaced y in their expansion for part (b) by $x^{0.5}$ and then integrated the resulting expression with respect to x . A common error amongst other solutions was to integrate the expansion from part (b) with respect to y , with some either leaving their answer in terms of y or others replacing y after the integration by $x^{0.5}$.

Candidates who didn't appreciate the link between the parts and expanded again often met problems with either the 'new' expansions or the powers involved.

Q17.

The majority of candidates did well on this question, with nearly half the candidates scoring 10 of the 12 marks or more.

In part (a) candidates were generally confident and competent in finding partial fractions and A and B were usually found correctly. The most common method was to substitute $x = -1$ and $x = -3/5$. The latter occasionally caused arithmetic problems. However, those who chose to use simultaneous equations or a mix of the two usual methods were just as successful.

In part (b) was pleasing to see that candidates could generally use the binomial expansions correctly and accurately and there were many complete correct solutions. The most common

errors occurred in the expansion of $(3 + 5x)^{-1}$. Instead of $3^{-1}\left(1 + \frac{5}{3}x\right)^{-1}$ some used $3\left(1 + \frac{5}{3}x\right)^{-1}$.

Some omitted the brackets around $\left(\frac{5x}{3}\right)^2$ too, although some continued correctly from here.

Almost all knew they were to combine the two series to get to the final result, although many sign and arithmetic errors were seen in attempting to do this. Very few misused the constants A and B found earlier, with just a few raising their result to these powers. Few candidates attempted this question by multiplying their expansions together.

In (c) many candidates showed they were not familiar with finding the range of valid values of x . Some just stated $-1 < x < 1$ while others assumed the answer to be $-5/3 < x < 5/3$. Some didn't even use an inequality but gave specific values for which they believed the expansions were not valid. Those who did find the correct limits often included the equality, so lost a mark.

Q18.

The majority of candidates answered part (a) successfully using the binomial expansion, Pascal's triangle or expanding brackets. Those candidates who wrote the expansion using the binomial coefficient notation from the formulae book did not always convince the examiner that they knew how to correctly evaluate the combinatorial coefficients.

In part (b)(i) very few candidates scored full marks with a large number of attempts seen where the numerator and denominator were integrated separately. Those candidates who

realized the need for negative indices generally scored well. Most candidates picked up the method mark in the final part of the question for dealing with the limits in a correct manner. A few candidates just stated the correct answer for the definite integral without showing any working. Such an approach gained no credit as it did not explicitly indicate the use of 'Hence'.

Q19.

In part (a)(i) about 75% of the candidates achieved both marks here. Some failed to achieve any marks through having the wrong sign on the second term, and some had the coefficient or sign wrong on the x^2 term.

In (a)(ii) most candidates knew they needed to take out a factor of 125, and of these most also

knew it was to the power $\frac{1}{3}$ thus demonstrating where the given 5 in the expression came from.

Those who failed to do this often just divided their expansion by 25 to obtain the given 5. Most

went on to gain the method mark of using $\pm \frac{27}{125}$ in their expression from part (a)(i) or starting the binomial expansion again. Although most got the coefficients of 9 and 81 numerically correct, there were often sign errors. These might have become apparent in part (b). Very few candidates attempted to use the binomial expansion formula from the formula book, and those that did use it often made a mistake in evaluating the coefficients.

In part (b) most candidates derived an appropriate value of x to use in their expression, but they couldn't achieve the expected answer from a wrong expression. Candidates should have expected to get a result close to 4.92 but this appeared not to have been considered by those who had an incorrect expansion, even by those who had a result

over 5. Some candidates just used their calculator to find $\sqrt[3]{119}$ for no credit, whilst a few thought they were to substitute $\sqrt[3]{119}$ into their expansion.

Q20.

This was a very straightforward question for candidates who were well versed in the necessary techniques. Even when sign errors and other small slips occurred, it was usually possible to obtain high marks. Most candidates knew that in part (b)(ii) they should mention the fact that h tends to zero, though it cannot actually be equal to zero for the question to make sense.

Q21.

Although the usual errors, which resulted in the wrong answers $1 + 7x + 21x^2 + 35x^3$ and $1 + 14x + 42x^2 + 70x^3$, were seen, there was some indication of an improvement in candidates' ability to apply the binomial expansion correctly. In part (b), the required method was generally well understood by average and more able candidates, but solutions with a correct coefficient of x^3 were a rarity, as the expansion of the squared term

$\left(1 - \frac{1}{2}x\right)^2$ was frequently incorrect.

Q22.

In part (a)(i), most candidates had the binomial expansion correct with some making an error in the x^2 term, with often the 2 being a 1.

Most candidates also had the expansion in part (a)(ii) correct. Most used the method of replacing x with $\frac{3}{4}x$, whilst some started the expansion again. The common error was again in the x^2 term, where the coefficient $\frac{3}{4}$ had not been squared.

In part (b), although some very neat, correct solutions were seen, most candidates could not handle the indices, although most knew they were to manipulate the expression to use the result from part (a)(ii). There were many errors in attempting to handle the indices $\pm \frac{1}{3}$. A few candidates who had handled the manipulation correctly, decided to double their final answer for no apparent reason.

Q23.

The sign of p was often awry, although that of q was usually correct. This was not penalised again in part (b)(i) where a lack of simplifying the coefficients and signs in the expressions was a greater source of mark loss. It was also disappointing to see a

significant minority of candidates integrating the term $+\frac{1}{x^6}$ instead of the printed term $-\frac{1}{x^6}$. However, in general, most candidates showed that they understood how to integrate x raised to a negative power.

Most knew how to evaluate a definite integral in part (b)(ii) but very few could maintain the numerical accuracy needed for the final mark. Those few candidates who just wrote down the value of the definite integral, presumably from a calculator, without showing how it had been obtained from the previous part (Hence) were awarded no credit. Some candidates, having correctly obtained '−1.7' for the value of the definite integral, went on to write 'Area = 1.7' as their final line of solution. This was not penalised and such candidates were awarded both marks.

Q24.

- (a) Virtually all candidates demonstrated confidence with the binomial theorem, and most completed part (a)(i) correctly, with only a few making either sign or coefficients errors, and then usually in the x^2 term. In part (a)(ii) most attempted the technique of taking out the factor 16, with many doing this correctly. The common errors were to just leave it as 16 rather than 64, or to leave 9 in the bracket rather

than $\frac{9}{16}$. Most candidates then used their result from part (a)(i) to attempt the expansion, whilst some started the binomial expansion again. Although some mistakes were made, the most common being omitting to

square the $\frac{9}{16}$, many candidates obtained the correct expansion. Those few candidates who attempted to use the expansion given in the formulae book were rarely successful, especially those who attempted to use nCr to find coefficients.

- (b) Most candidates who had the correct expansion in part (a), worked out that they needed to substitute $x = -\frac{1}{3}$ and obtained the correct values for a and b . Others who had an incorrect expansion in part (a) also often worked out that $x = -\frac{1}{3}$ was the value to substitute, but got a result nowhere near $13\frac{2}{3}$, and either didn't notice or just commented that it "didn't work". There was little evidence of checking back for an error. Those few candidates who proceeded by trial and improvement using a calculator gained no credit. Many candidates omitted part (b) altogether.

Q25.

There was a significant improvement from June 2008 in candidates' solutions to the binomial expansion topic. Part (a) was well answered, although a lack of brackets continues to be the main source of errors. Many candidates illustrated that they knew how to use their answer to part (a) to write down the expansion of $(1 - 2x)^4$, but not all solutions which finished with the printed answer were entirely convincing.

Part (c) was not always attempted and, even when attempted, the majority failed to convincingly show that there was only one stationary point. Many solutions included the correct equation ' $96x + 128x^3 = 0$ ', but a significant proportion of these lost the ' $x = 0$ ' root

and, by erroneously changing a sign, ended up with a value for x of the form $\frac{\sqrt{3}}{2}$.

Q26.

Many candidates confidently achieved full marks with this binomial expansion question, although some made sign errors or handled brackets incorrectly. Very few appeared to try to use the result given in the formula book. The infrequently-seen error in part (a)(i) was an incorrect sign in the x -term, although this had sometimes been corrected when it came

to part (a)(ii). In part (a)(ii), many candidates had problems restructuring $\sqrt{4 - x}$, often

getting it half right by leaving 4 rather than 2 on the front but with $\frac{x}{4}$ inside, or 2 on the front but with the inside left as x .

Those candidates who did have $\frac{x}{4}$ usually used it correctly in the expansion, with relatively few failing to square the 4 in the x^2 term. In part (b), most candidates realised

they were supposed to use $x = 1$, although some did try to use -1 or $\sqrt{3}$. Many candidates obtained results that clearly could not be $\sqrt{3}$, including some greater than 3, but this did not seem to suggest to them that they should look for an error. The correction of a fairly simple error might have given some candidates two further marks.

Q27.

Most candidates gave the correct value for n in part (a). The most common wrong answer was ' $n = -\frac{1}{4}$ '.

The expansion of $\left(1 + \frac{3}{x^2}\right)^2$ in part (b) caused more difficulty than expected. A common wrong answer was ' $1 + 6x^{-2} + 9x^4$ ' although there were many other examples of incorrect answers offered.

In integrating $\left(1 + \frac{3}{x^2}\right)^2$ in part (c) most candidates realised that their expansion in part (b) should be used but surprisingly the integration of 1 with respect to x caused just as many problems as the integration of the terms in x raised to negative powers. Although many candidates showed that they knew how to deal with the given limits, a lack of brackets in their numerical expressions frequently led to the wrong answer. The final mark was frequently lost because no **exact** answer (fraction or mixed number) was given.

Q28.

The unsimplified form of the binomial expansion in part (a) was usually correct. However in attempting to simplify their expressions many candidates made sign or coefficient errors, with answers such as $1 - x + x^2$ or $1 + x + 2x^2$. There were a pleasing number of correct expansions however. In part (b)(i) virtually all candidates showed they knew how to find partial fractions. Those who substituted $x = 1$ and $x = 2/3$ usually found the values of A and B correctly, with a few sign errors seen. Substitution of other values of x , or using simultaneous equations tended to be less successful.

There were many complete correct solutions to part (b)(ii) and most candidates demonstrated they knew in principle how to find the expansion. Many, though, made a mistake somewhere in their working. The most common error lay in the expansion of $(2 - 3x)^{-1}$. Many forgot to put 2 to the power -1 so the expansion of $(1 - 3x/2)^{-1}$ was often multiplied by 6 instead of $3/2$. Also some expanded $(1 - 3x)^{-1}$ instead of $(1 - 3x/2)^{-1}$. Some candidates forgot to multiply their separate expansions by their values of A and B or to add them together to complete the expansion.

The negative sign in $A = -2$ was often overlooked when combining the two expansions. Some candidates used the method of finding the two expansions and then finding the product of $(3x - 1)$ and two binomial expansions. This nearly always led to errors. Candidates should realise that there is often a link between parts of a question. The use of the partial fractions led to a more sensible method with far less algebra involved.

The majority of candidates scored no marks in part (c). Answers such as $x > 2/3$, $1 < x < 2/3$ abounded and there was confusion between $2/3$ and $3/2$. Many candidates apparently just made a guess, with x not equal to 1 or $2/3$ being a common one. Many gave an inequality in which a modulus was less than a negative number.

Q29.

Those candidates who used Pascal's triangle were generally able to find both expansions in part (a). Other candidates generally used the formulae booklet but some of these failed to display an understanding of the notation involved and were penalised heavily. In part (b), many candidates realised how to use their earlier expansions but the omission of brackets was a common error which led, for example, to the wrong answer " $1 + 12x + 12x^2 + 4x^3$ " in part (b)(i). Although some candidates did not attempt part (c), the vast majority used the correct method, although some careless arithmetic errors were seen.

Although the wording of the question has appeared in similar questions in past MPC2 papers (for example June 2005 Question (a)), there were a significant minority of candidates who just gave the answers (a)(i) $3x$ (ii) $4x$ (b)(i) $12x$ (ii) $12x$, but then many of these candidates, in their working for part (c), provided the full correct expansions required in part (b). In these cases the examiners awarded the marks for part (b) in part (c).

Q30.

Part (a) was generally done well by candidates, with very few errors seen in the binomial coefficients. Part (b) was similarly started well, and candidates either realised that they were to use their result from part (a) or simply started again. Errors were commonly made

in evaluating the coefficient of the x^2 term, often omitting to square $\frac{3}{2}$. Some candidates were also careless with the sign. Some candidates attempted to write down a result for part (b) with no working shown, so if they were incorrect they usually scored no marks.

The responses to part (c) were more varied: some candidates showed little knowledge of how to manipulate the expression, while those who knew that they were aiming to get a

multiple of the result from part (b) often failed to obtain the required multiple of $\frac{1}{2}$, many obtaining $\frac{1}{4}$, 2 or 4 instead.

Q31.

Although many candidates applied a correct method to find the expansion of $\left(1 + \frac{4}{x^2}\right)^3$, accuracy marks were sometimes lost because brackets were missing and the factor 4 was not squared when finding the value of q . Although a significant minority of candidates failed to use the expansion in part (a), despite the word 'Hence', most other candidates obtained the method mark in part (b), but it was surprising to find candidates having more difficulty integrating '1' than integrating the terms in x . Many candidates scored the method mark in part (b)(ii) but arithmetical errors were common in the subsequent

evaluation.

Q32.

In part (a)(i) nearly all candidates achieved the first mark, for $1 \pm \frac{1}{4}x$, and obtained the correct coefficient in the x^2 term, although many made a sign error.

Most candidates attempted to take out the 81 in part (a)(ii) and the majority did it correctly, with some forgetting to include the power. The approach to the expansion then varied

between substituting $\frac{16}{81}$ into the result from part (a)(i) or starting the expansion again, and often a mix of the two was seen. As the result was given, complete accuracy in the expression leading to the result was expected for full marks. This included the few who had chosen to use the expansion of $(a + b)^n$ from the formula book. There were many who made errors including sign errors, omitting brackets, and omitting x or x^2 in the expression.

Many candidates worked out that they needed to substitute $x = \frac{1}{16}$ in part (b) and evaluated correctly, a few not noting the request for 7 decimal places. Some candidates just found the 4th root of 80 direct from a calculator and gained no credit. Some candidates substituted the 4th root of 80 into the binomial expansion.

Q33.

This question on the binomial expansion was the worst-answered question on the paper. Although a majority of candidates showed some understanding of the topic, many of these candidates failed to include the correct powers of 2 in the second and third terms. More serious errors involved either using the first row of Pascal's triangle as 1, and so taking 1, 7, 21, 35 as the first four binomial coefficients for $(1 + X)^8$ or ignoring the coefficient of x and writing " $a = 8$, $b = 28$ and $c = 56$ ". Although some elegant solutions were presented, in general it was common to see those candidates who attempted part (b) either just writing the x^3 term from part (a) or trying to expand $(1 + 2.5x + x^2)^8$.

Q34.

It was pleasing to see in both parts (a) and (b) that most candidates were setting out their binomial expansions accurately and clearly. Most were fully correct in part (a). However, in part (b) there were again candidates who did not write down sufficient detail to convince they knew where the given result had come from. The common errors were to take the 8 out as factor and leave it as 8 and then just to divide the ensuing result by 4 to get the answer given, or to ignore the 8 completely. A few candidates used the expansion of $(ax + b)^n$ with $a = 8$, which can lead to the correct answer, but needed detail in the numerical evaluation of the coefficients to convince it had been done.

In part (c) most candidates substituted $x = \frac{1}{3}$ as expected and demonstrated the given result. Those few who did make a mistake apparently did not look for it; they just claimed

their answer was close to the given $\frac{599}{288}$. Those few candidates who did not show where

this fraction came from, but used a calculator to show it was approximately $\sqrt[3]{9}$, gained no credit.

Q35.

This question proved to be a good source of marks for many candidates although the

expansion in part (b) was quite frequently incorrect with $\left(\frac{2}{x}\right)^2$ being written as $\frac{4}{x}$ or as $\frac{2}{x^2}$.

Most candidates displayed good differentiation skills in part (c) and even more realised that the answer to part (d) required evaluation of their answer to part (c) when $x = 2$. Even though the majority of candidates knew how to find the equation of the normal it was very common to find a mark lost in the final part of the question because candidates were unable to rearrange their answer in to the form stated in the question.



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